
GENERAL RELATIVITY

Tutorial problem set 8, 30.10.2020, with solutions

■ PROBLEM 22 Killing vectors.

- (a) Show that the commutator of two Killing vectors is a Killing vector. Show that a linear combination with constant coefficients of two Killing vectors is a Killing vector.
- (b) Show that Killing vectors satisfy the following identities

$$(i) \quad \nabla_\mu \nabla_\nu K^\rho = R^\rho{}_{\nu\mu\sigma} K^\sigma, \quad (22.1)$$

$$(ii) \quad \nabla_\mu \nabla_\nu K^\mu = R_{\nu\mu} K^\mu, \quad (22.2)$$

$$(iii) \quad K^\alpha \nabla_\alpha R = 0. \quad (22.3)$$

Hints: For (i) first show that for any vector V^ρ

$$[\nabla_\mu, \nabla_\sigma] V_\rho = R_{\rho\nu\mu\sigma} V^\nu, \quad (22.4)$$

and make use of Killing's equation and a properties of the Riemann tensor. For (iii) make use of one of Bianchi identities from last time and the fact that the 2-tensor can always be decomposed into a symmetric and an antisymmetric part.

SOLUTION: (a) Let K and L be Killing vectors, which obey the Killing equations,

$$\nabla_{(\mu} K_{\nu)} = 0, \quad \nabla_{(\mu} L_{\nu)} = 0. \quad (22.5)$$

Then their commutator, $\Omega = [K, L]$ has components, $\Omega^\mu = K^\nu \partial_\nu L^\mu - L^\nu \partial_\nu K^\mu$. We ought to check that Ω satisfies the Killing equation,

$$\nabla_{(\mu} \Omega_{\nu)} \stackrel{?}{=} 0. \quad (22.6)$$

To show that, note first that $\Omega^\mu = K^\alpha \nabla_\alpha L^\mu - L^\alpha \nabla_\alpha K^\mu$, from which we conclude, $\Omega_\nu = K^\alpha \nabla_\alpha L_\nu - L^\alpha \nabla_\alpha K_\nu = \nabla_K L_\nu - \nabla_L K_\nu$. Upon inserting this into the Killing equation for Ω (22.6) one obtains three types of terms,

$$\nabla_{(\mu} \Omega_{\nu)} = (\nabla_{(\mu} K^\alpha) \nabla_{\alpha} L_{\nu)} - (\nabla_{(\mu} L^\alpha) \nabla_{\alpha} K_{\nu)} \quad (22.7)$$

$$+ K^\alpha [\nabla_\alpha, \nabla_{(\mu}] L_{\nu)} - L^\alpha [\nabla_\alpha, \nabla_{(\mu}] K_{\nu)} \quad (22.8)$$

$$+ K^\alpha \nabla_\alpha \nabla_{(\mu} L_{\nu)} - L^\alpha \nabla_\alpha \nabla_{(\mu} K_{\nu)}. \quad (22.9)$$

The third line (22.9) trivially vanishes. The first line (22.7) can be shown to vanish by making use of the Killing equations for K and L , $\nabla_\mu K^\alpha = -\nabla^\alpha K_\mu$ and $\nabla_\mu L^\alpha = -\nabla^\alpha L_\mu$. The second line (22.8) can be rewritten as,

$$\begin{aligned} & K^\alpha R_{\nu\beta\alpha\mu} L^\beta + K^\alpha R_{\mu\beta\alpha\nu} L^\beta - L^\alpha R_{\nu\beta\alpha\mu} K^\beta - L^\alpha R_{\mu\beta\alpha\nu} K^\beta \\ &= K^\alpha L^\beta [R_{\nu\beta\alpha\mu} + R_{\mu\beta\alpha\nu} - R_{\nu\alpha\beta\mu} - R_{\mu\alpha\beta\nu}] \\ &= K^\alpha L^\beta [R_{\nu\beta\alpha\mu} + R_{\mu\beta\alpha\nu} + R_{\nu\beta\mu\alpha} + R_{\nu\mu\alpha\beta} + R_{\mu\beta\nu\alpha} + R_{\mu\nu\alpha\beta}] = 0, \end{aligned} \quad (22.10)$$

where in the second equality we made use of the antisymmetry property of the Riemann tensor, $R_{\alpha[\beta\gamma\delta]} = 0$ and in the last equality we used, $R_{\alpha\beta\gamma\delta} = -R_{\alpha\beta\delta\gamma}$ and $R_{\alpha\beta\gamma\delta} = -R_{\delta\alpha\gamma\beta}$.

To show the second part, let $M = aK + bL$, with a, b real numbers, then $\nabla_{(\mu}M_{\nu)} = a\nabla_{(\mu}K_{\nu)} + b\nabla_{(\mu}L_{\nu)} = 0$.

(b) To get the first identity (22.1), note first that by covariance one can equivalently write it as,

$$\nabla_{\mu}\nabla_{\nu}K_{\rho} = R_{\rho\nu\mu\sigma}K^{\sigma}, \quad \nabla_{\mu}K_{\rho} = -\nabla_{\rho}K_{\mu}, \quad (22.11)$$

where for completeness we also wrote the Killing equation. The following sequence of identities then hold,

$$\begin{aligned} \nabla_{\mu}\nabla_{\nu}K_{\rho} &= \nabla_{\nu}\nabla_{\mu}K_{\rho} + [\nabla_{\mu}, \nabla_{\nu}]K_{\rho} = -\nabla_{\nu}\nabla_{\rho}K_{\mu} + [\nabla_{\mu}, \nabla_{\nu}]K_{\rho} \\ &= -[\nabla_{\nu}, \nabla_{\rho}]K_{\mu} - \nabla_{\rho}\nabla_{\nu}K_{\mu} + [\nabla_{\mu}, \nabla_{\nu}]K_{\rho} = -R_{\mu\sigma\nu\rho}K^{\sigma} - \nabla_{\rho}\nabla_{\nu}K_{\mu} + [\nabla_{\mu}, \nabla_{\nu}]K_{\rho} \\ &= R_{\rho\nu\mu\sigma}K^{\sigma} + [\nabla_{\mu}\nabla_{\nu}K_{\rho} + \nabla_{\nu}\nabla_{\rho}K_{\mu} + \nabla_{\rho}\nabla_{\mu}K_{\nu}], \end{aligned} \quad (22.12)$$

where to get the last equality we used symmetries of the Riemann tensor for the first term, and the Killing equation for the other terms. To prove (22.1) we need to show that the terms in square brackets vanish. Note first that, with the use of the Killing equation these terms can be reorganized as the antisymmetrized sum over all three indices,

$$\begin{aligned} 3\nabla_{[\mu}\nabla_{\nu}K_{\rho]} &= \frac{1}{2}\{\nabla_{\mu}\nabla_{\nu}K_{\rho} + \nabla_{\nu}\nabla_{\rho}K_{\mu} + \nabla_{\rho}\nabla_{\mu}K_{\nu} - \nabla_{\mu}\nabla_{\rho}K_{\nu} - \nabla_{\nu}\nabla_{\mu}K_{\rho} - \nabla_{\rho}\nabla_{\nu}K_{\mu}\} \\ &= \frac{1}{2}\{[\nabla_{\mu}, \nabla_{\nu}]K_{\rho} + [\nabla_{\nu}, \nabla_{\rho}]K_{\mu} + [\nabla_{\rho}, \nabla_{\mu}]K_{\nu}\} \\ &= \frac{1}{2}\{R_{\rho\sigma\mu\nu} + R_{\mu\sigma\nu\rho} + R_{\nu\sigma\rho\mu}\}K^{\sigma} = -\frac{1}{2}\{R_{\sigma\rho\mu\nu} + R_{\sigma\mu\nu\rho} + R_{\sigma\nu\rho\mu}\}K^{\sigma} \\ &= -\frac{3}{2}R_{\sigma[\rho\mu\nu]}K^{\sigma} = 0, \end{aligned} \quad (22.13)$$

because of the symmetry of the Riemann tensor. We have thus shown that,

$$\nabla_{\mu}\nabla_{\nu}K_{\rho} = R_{\rho\nu\mu\sigma}K^{\sigma}. \quad (22.14)$$

This then completes the proof of (22.1).

To show (22.2) act $g^{\mu\rho}$ on (22.1) to obtain,

$$\nabla_{\mu}\nabla_{\nu}K^{\mu} = R_{\nu\mu}K^{\mu}. \quad (22.15)$$

To prove the last identity (22.3), let us rewrite (22.15) as,

$$\square K_{\nu} = -R_{\nu\mu}K^{\mu}, \quad (22.16)$$

where $\square = g^{\alpha\beta}\nabla_{\alpha}\nabla_{\beta}$ is the D'Alembert operator. Next, acting ∇^{ν} on (22.16), gives,

$$\nabla^{\nu}\square K_{\nu} = -\nabla^{\nu}R_{\nu\mu}K^{\mu} = -\frac{1}{2}K^{\mu}\nabla_{\mu}R, \quad (22.17)$$

where in the last step we used the second Bianchi identity. What remains to be done is to show that the left hand side of Eq. (22.17) vanishes. Since the D'Alembertian is a scalar operator, the derivative commutes with it, and the left hand side of (22.17) gives,

$$\nabla^{\nu}\square K_{\nu} = \square(\nabla^{\nu}K_{\nu}) = \square(g^{\mu\nu}\nabla_{\mu}K_{\nu}) = \square(g^{\mu\nu}\nabla_{(\mu}K_{\nu)}) = 0, \quad (22.18)$$

completing the proof of (22.3). This means that the Ricci curvature scalar is conserved by parallel transport along any curve generated by a Killing vector (*i.e.* whose tangent vector is given by the Killing vector).

■ **PROBLEM 23** Killing vectors on a 2-sphere.

The two-sphere is given by

$$ds^2 = d\vartheta^2 + \sin^2(\vartheta) d\varphi^2 . \quad (23.1)$$

(a) Solve the Killing's equation to find that the Killing vectors of a 2-sphere are

$$K_1 = \cos(\varphi) \partial_{\vartheta} - \cot(\vartheta) \sin(\varphi) \partial_{\varphi} , \quad (23.2)$$

$$K_2 = -\sin(\varphi) \partial_{\vartheta} - \cot(\vartheta) \cos(\varphi) \partial_{\varphi} , \quad (23.3)$$

$$K_3 = \partial_{\varphi} . \quad (23.4)$$

(b) Show that the commutators of these three Killing vectors satisfy the following commutation relations

$$[K_1, K_2] = K_3 , \quad (23.5)$$

$$[K_2, K_3] = K_1 , \quad (23.6)$$

$$[K_3, K_1] = K_2 . \quad (23.7)$$

Check that they satisfy the algebra of Lie groups. Can you see which algebra is it?

SOLUTION: (a) Making use of the nonvanishing Christoffel symbols for the sphere (23.1),

$$\Gamma_{\varphi\varphi}^{\vartheta} = -\sin(\vartheta) \cos(\vartheta) , \quad \Gamma_{\vartheta\varphi}^{\varphi} = \Gamma_{\varphi\vartheta}^{\varphi} = \cot(\vartheta) , \quad (23.8)$$

one obtains the following Killing equations,

$$\nabla_{(\vartheta} K_{\vartheta)} = \partial_{\vartheta} K_{\vartheta} = 0 \quad (23.9)$$

$$2\nabla_{(\vartheta} K_{\varphi)} = \partial_{\vartheta} K_{\varphi} + \partial_{\varphi} K_{\vartheta} - 2 \cot(\vartheta) K_{\varphi} = 0 \quad (23.10)$$

$$\nabla_{(\varphi} K_{\varphi)} = \partial_{\varphi} K_{\varphi} + \sin(\vartheta) \cos(\vartheta) K_{\vartheta} = 0 . \quad (23.11)$$

From the first equation we see that $K_{\vartheta} = K_{\vartheta}(\varphi)$. We shall try a separation of variables *Ansatz*,

$$K_{\vartheta} = \Psi(\varphi) , \quad K_{\varphi} = \Theta(\vartheta) \Phi(\varphi) . \quad (23.12)$$

Inserting this into (23.10) and (23.11) gives,

$$\Phi(\varphi) \left[\frac{d\Theta(\vartheta)}{d\vartheta} - 2 \cot(\vartheta) \Theta(\vartheta) \right] = -\frac{d\Psi(\varphi)}{d\varphi} = C_1 \Phi(\varphi) \quad (23.13)$$

$$\Theta(\vartheta) \left[\frac{d\Phi(\varphi)}{d\varphi} \right] = -[\sin(\vartheta) \cos(\vartheta)] \Psi(\varphi) = C_2 \Theta(\vartheta) \Psi(\varphi) , \quad (23.14)$$

where C_1 and C_2 are constants (each of which may, but need not, vanish). This is telling us that, if the separation of variable *Ansatz* is correct, then the problem separates into four ordinary differential equations. The equations for Φ are,

$$\frac{d\Psi(\varphi)}{d\varphi} = -C_1 \Phi(\varphi) , \quad \frac{d\Phi(\varphi)}{d\varphi} = C_2 \Psi(\varphi) \quad (23.15)$$

and the ones for Θ are,

$$\Theta(\vartheta) = -\frac{1}{C_2} \sin(\vartheta) \cos(\vartheta) , \quad \frac{d\Theta(\vartheta)}{d\vartheta} - 2 \cot(\vartheta) \Theta(\vartheta) = C_1 , \quad (23.16)$$

The equations for Θ are in general inconsistent. Let us ignore that for a moment, and look at (23.15). General solutions are harmonic functions of the form,

$$\Psi(\varphi) = A \cos(n\varphi) + B \sin(n\varphi), \quad \Phi(\varphi) = D \cos(n\varphi) + E \sin(n\varphi). \quad (23.17)$$

where we made use of the fact that $\Omega = \sqrt{C_1 C_2} = n$ must be a positive integer (because it must be a periodic function on S^2). Not all of the constants are free to choose. Indeed the first order equations dictate the following relations, $-nA = -C_1 E$, $nB = -C_1 D$, $-nD = C_2 B$, $nE = C_2 A$. This basically means that the following pairs of solutions are consistent (up to normalization constants yet to be determined),

$$(\Psi, \Phi) \sim (\cos(n\varphi), \sin(n\varphi)), \quad (\Psi, \Phi) \sim (\sin(n\varphi), \cos(n\varphi)). \quad (23.18)$$

However, not all of these solutions are consistent with the equations for Θ in (23.16).

The strategy of constructing the solutions is now to select a harmonic solution from (23.18) and check whether the principal equations (23.13–23.14) can be solved.

Take for example $n = 0$, for which $(\Psi, \Phi) \sim (1, 0)$ or $(0, 1)$, Inserting these into (23.13–23.14) we see that one the second pair can satisfy both equations, provided one chooses $C_1 = 0$ and $\Theta = \sin^2(\vartheta)$. Putting everything together, we have a first Killing vector, $(K_3)_\mu = (0, \sin^2(\vartheta))$, which implies $(K_3)^\mu = (0, 1)$, or equivalently $K_3 = \partial_\varphi$, where our notation follows Eq. (23.4).

For $n = 1$ we have the following two pairs, $(\Psi, \Phi) \sim (\cos(\varphi), \sin(\varphi))$ or $(\Psi, \Phi) \sim (\sin(\varphi), \cos(\varphi))$.

Let us first consider $(\Psi, \Phi) \sim (\cos(\varphi), \sin(\varphi))$. A convenient choice is $C_1 = C_2 = 1$, such that the first order equations (23.15) give (up to an unimportant normalization constant),

$$\Psi = \cos(\varphi), \quad \Phi = \sin(\varphi). \quad (23.19)$$

Inserting these into (23.13–23.14) results in,

$$\sin(\varphi) \left[\frac{d\Theta(\vartheta)}{d\vartheta} - 2 \cot(\vartheta) \Theta(\vartheta) \right] = \sin(\varphi) = \sin(\varphi) \quad (23.20)$$

$$\Theta(\vartheta) \cos(\varphi) = -[\sin(\vartheta) \cos(\vartheta)] \cos(\varphi) = \Theta(\vartheta) \cos(\varphi). \quad (23.21)$$

A consistent solution of both equations exists, and it is given by, $\Theta = -\sin(\vartheta) \cos(\vartheta)$, which then implies for the second Killing vector, $(K_1)_\mu = (\cos(\varphi), -\sin(\vartheta) \cos(\vartheta) \sin(\varphi))$, from which we conclude, $K_1^\mu = (\cos(\varphi), -\cot(\vartheta) \sin(\varphi))$, or equivalently, $K_1 = \cos(\varphi) \partial_\vartheta - \cot(\vartheta) \sin(\varphi) \partial_\varphi$.

Next, we consider the second pair of harmonic functions, $(\Psi, \Phi) \sim (\sin(\varphi), \cos(\varphi))$. The first order equations fix the relative sign (23.15), we we have (with $C_1 = 1 = C_2$ and setting the normalization constant to minus one - an overall normalization constant is not important),

$$\Psi = -\sin(\varphi), \quad \Phi = \cos(\varphi). \quad (23.22)$$

Inserting these into (23.13–23.14) results in,

$$\cos(\varphi) \left[\frac{d\Theta(\vartheta)}{d\vartheta} - 2 \cot(\vartheta) \Theta(\vartheta) \right] = \cos(\varphi) = \cos(\varphi) \quad (23.23)$$

$$\Theta(\vartheta) [-\sin(\varphi)] = [\sin(\vartheta) \cos(\vartheta)] \sin(\varphi) = -\Theta(\vartheta) \sin(\varphi). \quad (23.24)$$

Just as in the previous case, a consistent solution of both equations exists, and it is given by, $\Theta = -\sin(\vartheta) \cos(\vartheta)$, which then implies for the second Killing vector, $(K_2)_\mu = (-\sin(\varphi), -\sin(\vartheta) \cos(\vartheta) \cos(\varphi))$, from which we conclude, $K_2^\mu = (-\sin(\varphi), -\cot(\vartheta) \cos(\varphi))$, or equivalently, $K_2 = -\sin(\varphi) \partial_\vartheta - \cot(\vartheta) \cos(\varphi) \partial_\varphi$.

Of course, all of the above Killing vectors are determined up to a normalization constant, which was chosen in the above analysis to reproduce the Killing vectors given above in (23.4).

What remains to check whether other harmonic functions produce additional Killing vectors. To probe that, let us consider the pair $(\Psi, \Phi) \sim (\sin(n\varphi), \cos(n\varphi))$ and make a convenient choice $C_1 = n = C_2$ (one can show that the result does not depend on that choice) from

$$\Psi = -\sin(n\varphi), \quad \Phi = \cos(n\varphi), \quad (n \geq 2). \quad (23.25)$$

Inserting these into (23.13–23.14) results in,

$$\cos(n\varphi) \left[\frac{d\Theta(\vartheta)}{d\vartheta} - 2 \cot(\vartheta) \Theta(\vartheta) \right] = n \cos(n\varphi) = n \cos(\varphi) \quad (23.26)$$

$$\Theta(\vartheta) [-n \sin(n\varphi)] = [\sin(\vartheta) \cos(\vartheta)] \sin(n\varphi) = -n \Theta(\vartheta) \sin(n\varphi). \quad (23.27)$$

From the second equation we conclude,

$$\Theta(\vartheta) = -\frac{\sin(\vartheta) \cos(\vartheta)}{n}. \quad (23.28)$$

Consistency of the first equation (23.26) demands

$$\frac{d\Theta(\vartheta)}{d\vartheta} - 2 \cot(\vartheta) \Theta(\vartheta) = n \quad (23.29)$$

which is solved for $n = 1$, but not for $n \geq 2$, completing the construction of the Killing vectors for the sphere.

Of course, all of the above Killing vectors are determined up to a normalization constant, which was chosen in the above analysis to reproduce the Killing vectors given above in (23.4) by explicitly solving the Killing equations. There is a much easier way of getting the Killing vectors for the sphere. One can embed the sphere into the three dimensional (flat) Euclidean space, whose coordinates are (X, Y, Z) , with the embedding constraint, $X^2 + Y^2 + Z^2 = 1$. The Killing vectors are rotations in the XY , YZ and ZX planes (translations are excluded by the embedding constraint). Transforming the rotational killing vectors of the embedding space into the spherical coordinates on the sphere gives (23.4).

(b) One can calculate the desired commutation relations (23.7) by making use of the known formula, $[K_i, K_j]^\mu = K_i^\alpha \partial_\alpha K_j^\mu - K_j^\alpha \partial_\alpha K_i^\mu$.

Thus we have,

$$[K_1, K_2]^\vartheta = [-\cot(\vartheta) \sin(\varphi)] \partial_\varphi [-\sin(\varphi)] - [-\cot(\vartheta) \cos(\varphi)] \partial_\varphi [\cos(\varphi)] = 0 \quad (23.30)$$

$$\begin{aligned} [K_1, K_2]^\varphi &= [\cos(\varphi) \partial_\vartheta - \cot(\vartheta) \sin(\varphi) \partial_\varphi] [-\cot(\vartheta) \cos(\varphi)] \\ &\quad - [-\sin(\varphi) \partial_\vartheta - \cot(\vartheta) \cos(\varphi) \partial_\varphi] [-\cot(\vartheta) \sin(\varphi)] = 1. \end{aligned} \quad (23.31)$$

This means that the vector field corresponding to the commutator $[K_1, K_2]$ has the components, $[K_1, K_2]^\mu = (0, 1)$, or when written in the differential basis, $[K_1, K_2] = \partial_\varphi$, which is equal to K_3 .

The other two commutators are evaluated in a similar fashion. Firstly, we have $[K_2, K_3]$,

$$[K_2, K_3]^\vartheta = -\partial_\varphi [-\sin(\varphi)] = \cos(\varphi) \quad (23.32)$$

$$[K_2, K_3]^\varphi = -\partial_\varphi [-\cot(\vartheta) \cos(\varphi)] = -\cot(\vartheta) \sin(\varphi), \quad (23.33)$$

that is, $[K_2, K_3]^\mu = (\cos(\varphi), -\cot(\vartheta)\sin(\varphi))$. When written in the differential basis, $[K_2, K_3] = \cos(\varphi)\partial_\vartheta - \cot(\vartheta)\sin(\varphi)\partial_\varphi = K_1$. Finally, we have for $[K_3, K_1]$,

$$[K_3, K_1]^\vartheta = \partial_\varphi[\cos(\varphi)] = -\sin(\varphi) \quad (23.34)$$

$$[K_3, K_1]^\varphi = \partial_\varphi[-\cot(\vartheta)\sin(\varphi)] = -\cot(\vartheta)\cos(\varphi), \quad (23.35)$$

or $[K_3, K_1]^\mu = (-\sin(\varphi), -\cot(\vartheta)\cos(\varphi)) = K_2^\mu$, *i.e.*

$$[K_3, K_1] = -\sin(\varphi)\partial_\vartheta - \cot(\vartheta)\cos(\varphi)\partial_\varphi = K_2.$$

Putting everything together, reproduces the algebra (23.7). This is the Lie algebra of special rotations (special refers to the determinant equal to unity) in three spatial dimensions, $so(3)$. The corresponding group is $SO(3)$, and it can be obtained by exponentiating the algebra, *i.e.* if $g \in SO(3)$ and if $a \in so(3)$, then $g = \exp(a)$, where $a = \sum_{i=1}^3 a_i K_i$ ($a_i \in \mathbf{R}$) is an arbitrary element in the algebra.

A related algebra is the algebra of Pauli matrices,

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (23.36)$$

which corresponds to the algebra $su(2)$, which is the algebra of the special unitary group $SU(2)$ in two (complex) dimensions. Pauli matrices satisfy, $[\sigma^i, \sigma^j] = 2i\epsilon^{ijl}\sigma^l$ ($i, j, l = 1, 2, 3$), such that $\sigma^i/2$ gives the algebra with properly normalized structure constants, $f^{ijl} = i\epsilon^{ijl}$. Because a simple rescaling, $-i\sigma^j/2 \rightarrow K_i$ maps one algebra onto the other, the two algebras have the same structure, *i.e.* the corresponding groups are locally the same (there is a homeomorphism, or even a diffeomorphism between the two). However, globally (topologically) they differ. Do you know how they differ globally?

■ PROBLEM 24 Maximally symmetric spaces.

The metric of a *de Sitter space* is given by

$$ds^2 = \frac{1}{\cos^2(H\eta)} \left[-d\eta^2 + \frac{d\chi^2}{H^2} + \frac{\sin^2(\chi)}{H^2} (d\vartheta^2 + \sin^2(\vartheta)d\varphi^2) \right] \quad (24.1)$$

(a) Calculate the Riemann tensor, the Ricci tensor and the Ricci scalar.

Note: For a diagonal metric Christoffel symbols can be calculated using the following simplified formulas

$$\begin{aligned} \Gamma_{\alpha\beta}^\gamma &= 0, & \Gamma_{\alpha\alpha}^\beta &= -\frac{1}{2g_{\beta\beta}}\partial_\beta g_{\alpha\alpha}, \\ \Gamma_{\alpha\beta}^\beta &= \partial_\alpha \left(\ln \sqrt{|g_{\beta\beta}|} \right), & \Gamma_{\alpha\alpha}^\alpha &= \partial_\alpha \left(\ln \sqrt{|g_{\alpha\alpha}|} \right), \end{aligned} \quad (24.2)$$

where $\alpha \neq \beta \neq \gamma \neq \alpha$ and there is *no* summation over indices.

(b) Show that

$$R_{\rho\sigma\mu\nu} = \frac{R}{D(D-1)}(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}), \quad (D=4). \quad (24.3)$$

A space for which this last equation is satisfied is called a maximally symmetric space. In such a space, the curvature is the same everywhere and the same in every direction. Hence, if one knows the curvature in one point of the space, it is known everywhere.

SOLUTION: (a) Firstly, the nonvanishing Christoffel symbols are,

$$\begin{aligned}\Gamma_{\eta\eta}^\eta &= H \tan(H\eta), & \Gamma_{\chi\chi}^\eta &= \frac{\tan(H\eta)}{H}, & \Gamma_{\vartheta\vartheta}^\eta &= \frac{\tan(H\eta)}{H} \sin^2(\chi), & \Gamma_{\varphi\varphi}^\eta &= \frac{\tan(H\eta)}{H} \sin^2(\chi) \sin^2(\vartheta) \\ \Gamma_{\eta\chi}^\chi &= H \tan(H\eta), & \Gamma_{\vartheta\vartheta}^\chi &= -\sin(\chi) \cos(\chi), & \Gamma_{\varphi\varphi}^\chi &= -\sin(\chi) \cos(\chi) \sin^2(\vartheta), \\ \Gamma_{\eta\vartheta}^\vartheta &= H \tan(H\eta), & \Gamma_{\vartheta\chi}^\vartheta &= \cot(\chi), & \Gamma_{\varphi\varphi}^\vartheta &= -\sin(\vartheta) \cos(\vartheta), \\ \Gamma_{\eta\varphi}^\varphi &= H \tan(H\eta), & \Gamma_{\varphi\chi}^\varphi &= \cot(\chi), & \Gamma_{\varphi\vartheta}^\varphi &= \cot(\vartheta),\end{aligned}\quad (24.4)$$

From here we can construct the nonvanishing elements of the Riemann tensor. Making use of the definition,

$$R_{\beta\mu\nu}^\alpha = \partial_\mu \Gamma_{\nu\beta}^\alpha + \Gamma_{\mu\sigma}^\alpha \Gamma_{\nu\beta}^\sigma - (\mu \leftrightarrow \nu) \quad (24.5)$$

we can write, for example, for the $\eta\chi\eta\chi$ -component,

$$R_{\chi\eta\chi}^\eta = \partial_\eta \Gamma_{\chi\chi}^\eta + \Gamma_{\eta\sigma}^\eta \Gamma_{\chi\chi}^\sigma - \partial_\chi \Gamma_{\eta\chi}^\eta - \Gamma_{\chi\sigma}^\eta \Gamma_{\eta\chi}^\sigma \quad (24.6)$$

Note that $\sigma = \eta$ in the first product of Christoffel symbols and $\sigma = \chi$ contributes in the second product of Christoffel symbols. Evaluating (24.6) gives,

$$R_{\chi\eta\chi}^\eta = \frac{1}{\cos^2(H\eta)}, \quad (24.7)$$

such that,

$$R_{\eta\chi\eta\chi} = -\frac{1}{\cos^4(H\eta)} = H^2 [g_{\eta\eta} g_{\chi\chi}] = -R_{\chi\eta\eta\chi} = -R_{\eta\chi\chi\eta} = R_{\chi\eta\chi\eta}, \quad (24.8)$$

where to get the subsequent equalities we made use of the symmetries of the Riemann tensor.

Analogously, one obtains the other non-vanishing components of the Riemann tensor,

$$R_{\vartheta\eta\vartheta}^\eta = \partial_\eta \Gamma_{\vartheta\vartheta}^\eta + \Gamma_{\eta\sigma}^\eta \Gamma_{\vartheta\vartheta}^\sigma - \partial_\vartheta \Gamma_{\eta\vartheta}^\eta - \Gamma_{\vartheta\sigma}^\eta \Gamma_{\eta\vartheta}^\sigma = \frac{\sin^2(\chi)}{\cos^2(H\eta)} \quad (24.9)$$

$$R_{\varphi\eta\varphi}^\eta = \partial_\eta \Gamma_{\varphi\varphi}^\eta + \Gamma_{\eta\sigma}^\eta \Gamma_{\varphi\varphi}^\sigma - \partial_\varphi \Gamma_{\eta\varphi}^\eta - \Gamma_{\varphi\sigma}^\eta \Gamma_{\eta\varphi}^\sigma = \frac{\sin^2(\chi) \sin^2(\vartheta)}{\cos^2(H\eta)} \quad (24.10)$$

$$R_{\vartheta\chi\vartheta}^\chi = \partial_\chi \Gamma_{\vartheta\vartheta}^\chi + \Gamma_{\chi\sigma}^\chi \Gamma_{\vartheta\vartheta}^\sigma - \partial_\vartheta \Gamma_{\chi\vartheta}^\chi - \Gamma_{\vartheta\sigma}^\chi \Gamma_{\chi\vartheta}^\sigma = \frac{\sin^2(\chi)}{\cos^2(H\eta)} \quad (24.11)$$

$$R_{\varphi\chi\varphi}^\chi = \partial_\chi \Gamma_{\varphi\varphi}^\chi + \Gamma_{\chi\sigma}^\chi \Gamma_{\varphi\varphi}^\sigma - \partial_\varphi \Gamma_{\chi\varphi}^\chi - \Gamma_{\varphi\sigma}^\chi \Gamma_{\chi\varphi}^\sigma = \frac{\sin^2(\chi) \sin^2(\vartheta)}{\cos^2(H\eta)} \quad (24.12)$$

$$R_{\varphi\vartheta\varphi}^\vartheta = \partial_\vartheta \Gamma_{\varphi\varphi}^\vartheta + \Gamma_{\vartheta\sigma}^\vartheta \Gamma_{\varphi\varphi}^\sigma - \partial_\varphi \Gamma_{\vartheta\varphi}^\vartheta - \Gamma_{\varphi\sigma}^\vartheta \Gamma_{\vartheta\varphi}^\sigma = \frac{\sin^2(\chi) \sin^2(\vartheta)}{\cos^2(H\eta)}, \quad (24.13)$$

such that

$$R_{\eta\vartheta\eta\vartheta} = -R_{\vartheta\eta\eta\vartheta} = -R_{\eta\vartheta\vartheta\eta} = R_{\vartheta\eta\vartheta\eta} = -\frac{\sin^2(\chi)}{\cos^4(H\eta)} = H^2 [g_{\eta\eta} g_{\vartheta\vartheta}], \quad (24.14)$$

$$R_{\eta\varphi\eta\varphi} = -R_{\varphi\eta\eta\varphi} = -R_{\eta\varphi\varphi\eta} = R_{\varphi\eta\varphi\eta} = -\frac{\sin^2(\chi) \sin^2(\vartheta)}{\cos^4(H\eta)} = H^2 [g_{\eta\eta} g_{\varphi\varphi}], \quad (24.15)$$

$$R_{\chi\vartheta\chi\vartheta} = -R_{\vartheta\chi\chi\vartheta} = -R_{\chi\vartheta\vartheta\chi} = R_{\vartheta\chi\vartheta\chi} = \frac{\sin^2(\chi)}{H^2 \cos^4(H\eta)} = [g_{\chi\chi} g_{\vartheta\vartheta}], \quad (24.16)$$

$$R_{\chi\varphi\chi\varphi} = -R_{\varphi\chi\chi\varphi} = -R_{\chi\varphi\varphi\chi} = R_{\varphi\chi\varphi\chi} = \frac{\sin^2(\chi) \sin^2(\vartheta)}{H^2 \cos^4(H\eta)} = H^2 [g_{\chi\chi} g_{\varphi\varphi}], \quad (24.17)$$

$$R_{\vartheta\varphi\vartheta\varphi} = -R_{\varphi\vartheta\vartheta\varphi} = -R_{\vartheta\varphi\varphi\vartheta} = R_{\varphi\vartheta\varphi\vartheta} = \frac{\sin^4(\chi) \sin^2(\vartheta)}{H^2 \cos^4(H\eta)} = H^2 [g_{\vartheta\vartheta} g_{\varphi\varphi}]. \quad (24.18)$$

Note that the component $R_{\eta\chi\eta}^\chi$ need not be calculated separately, since it can be obtained from the symmetries of the Riemann tensor and already calculated components, $R_{\eta\chi\eta}^\chi = g^{\chi\chi} R_{\chi\eta\chi\eta} = -H^2/\cos^2(H\eta)$. Note also that the Riemann tensor vanishes when all four indices are different (this is so because all Christoffel symbols vanish if all three indices are different). Also, if three indices are different and one pair is the same, then the Riemann tensor vanishes.

(b) It is easily checked that all of the above results (24.14–24.18) fit the pattern in equation (24.3), which in $D = 4$ reduces to,

$$R_{\alpha\beta\mu\nu} = H^2 [g_{\alpha\mu}g_{\beta\nu} - g_{\alpha\nu}g_{\beta\mu}] . \quad (24.19)$$